

Problem-Solution Fit

Date	5th July 2026
Team ID	RAHMA HUSSEIN JUMA
Project Name	CREDIT CARD APPROVAL PREDICTION
Maximum Marks	5 Marks

Problem-Solution Fit Canvas: .

1. CUSTOMER SEGMENT(S) [CS] Bank credit analysts, financial compliance officers, and prospective customers applying for a credit card.	6. CUSTOMER LIMITATIONS [CL] Analysts have limited time per application and no easy way to cross-reference full credit history at a glance. Customers have no visibility into their own approval odds before applying. Compliance officers work with confusing multi-class payment status codes that aren't immediately readable.	5. AVAILABLE SOLUTIONS PROS & CONS [AS] • <i>Manual review</i> — Pros: full human judgment; Cons: slow, inconsistent, error-prone at scale • <i>Simple rule-based scoring (e.g., income cutoffs)</i> — Pros: easy to implement; Cons: too rigid, misses nuanced risk patterns • <i>Generic credit bureau scores</i> — Pros: industry standard; Cons: not tailored to this bank's own historical applicant data
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2. PROBLEMS / PAINS

+ ITS FREQUENCY [PR]

Thousands of applications arrive daily; each one takes significant manual time to assess, and mistakes happen regularly under that volume — this is a constant, everyday pain, not an occasional one.

9. PROBLEM ROOT /

CAUSE [RC]

There is no fast, consistent, data-driven way to translate an applicant's financial and credit history into a reliable risk decision — every analyst effectively makes a judgment call using incomplete mental shortcuts instead of a systematic model.

7. BEHAVIOR

+ ITS INTENSITY [BE]

Analysts manually cross-check income, employment, and payment history for every single application, spending real time and mental effort per case — a moderate-to-high intensity behavior repeated at high frequency.

3. TRIGGERS TO ACT

[TR]

A new application arriving in the queue; a compliance audit flagging a batch of risky accounts; a customer wanting to know their eligibility before formally applying.

10. YOUR SOLUTION

[SL]

An ML-powered web application that takes an applicant's financial and demographic details, runs them through a trained classification model (best of four compared: Logistic Regression, Decision Tree, Random Forest, XGBoost), and instantly returns an Approved/Rejected prediction with a risk category — replacing slow manual review with a fast, consistent, data-driven decision.

8. CHANNELS OF BEHAVIOR [CH]

ONLINE

- Internal banking software used by analysts to log and review applications
- Applicant self-service web portal (our Flask app) for customers checking eligibility

4. EMOTIONS

BEFORE / AFTER [EM]

- *Before:* overwhelmed, anxious about making a wrong call, frustrated by slow manual review
- *After (with our solution):* confident, faster, reassured by a consistent, data-backed decision

OFFLINE

- Paper application forms collected at bank branches, later entered into the system by staff
- In-person document verification (proof of income, ID, employment letters) during branch visits